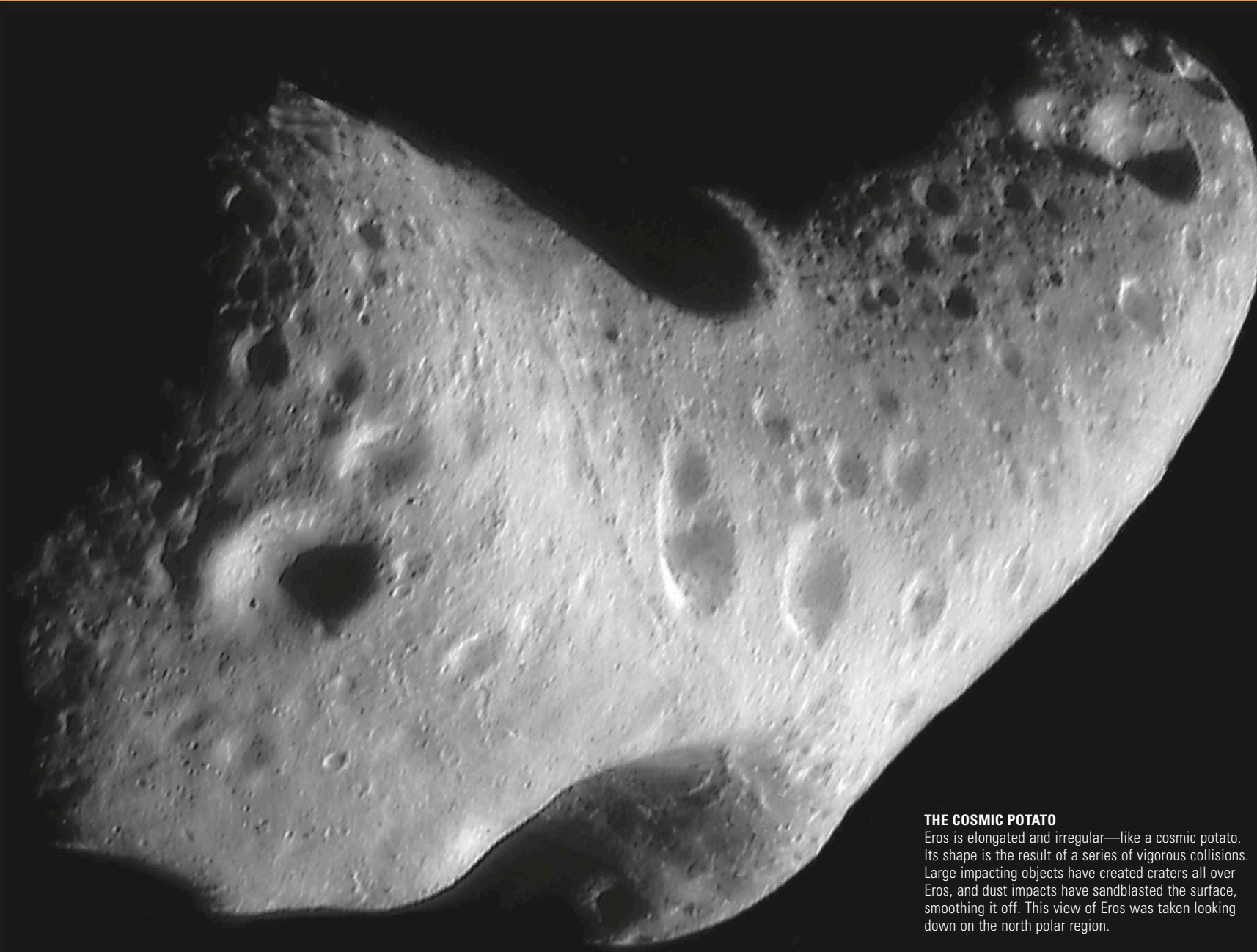




THE COSMIC POTATO

Eros is elongated and irregular—like a cosmic potato. Its shape is the result of a series of vigorous collisions. Large impacting objects have created craters all over Eros, and dust impacts have sandblasted the surface, smoothing it off. This view of Eros was taken looking down on the north polar region.

NEAR-EARTH ASTEROID

162173 Ryugu

AVERAGE DISTANCE TO SUN 111 million miles (178 million km)

ORBITAL PERIOD 1.30 years

ROTATION PERIOD 7.63 hours

DIAMETER 0.54 miles (0.87 km)

DATE OF DISCOVERY May 10, 1999

Ryugu is the target of the Japanese probe Hayabusa 2, which arrived in orbit around it in June 2018. After collecting samples, Hayabusa 2 left for Earth in November 2019 and is expected to return samples to Earth in 2020.

Ryugu has about twice the diameter of Bennu (see opposite) and also is likely to be a carbonaceous asteroid. It has a low density and a “spinning top” shape, which is controlled by its rapid rotation. Interestingly, both Ryugu and Bennu rotate in a retrograde sense, meaning that they spin clockwise about their axes while orbiting counterclockwise around the Sun.

Ryugu’s surface is strewn with large boulders, separated by smoother areas. Those are not so dusty as they might appear; seen up close, they consist mostly of angular rock fragments. Hayabusa 2 deployed four small hopping rovers to collect surface material. Some of the samples were freshly exposed from below the surface in a 33-ft (10-m) crater made by using a free-flying gun called the “Small Carry-on Impactor”, or SCI, to fire a copper projectile at the surface from a range of about 1,641 ft (500 m).

Ryugu’s orbit crosses the Earth’s, but it poses less of a hazard than Bennu because currently it is predicted not to come closer than about a quarter of the distance between the Moon and the Earth.

RYUGU

Strewn with angular boulders of all scales, Ryugu’s “spinning top” shape is clearly seen. The ridge along its equator bears at least one impact crater whose indistinct form may be symptomatic of this asteroid’s rubbly interior.

